

Decomposition and Distributed Algorithms for Home Healthcare Routing and Scheduling Problem

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Abstract—Many people in need of care still live in their homes, requiring the caretakers to travel to them. Assigning the people to caretakers and generating their schedules can be formulated as a mixed integer linear programming problem (MILP) that inherits many features of the well-known vehicle routing problem with time windows (VRPTW). Currently, the most successful exact algorithms for VRPTW are based on branch and price framework, which combine column generation (CG) and branching. While these methods could be successful for Home Healthcare Routing and Scheduling Problem (HHCRSP) as well, fast approximate algorithms are appealing, especially for large problems. We recently employed a heuristic distributed gossip algorithm to solve HHCRSP. The method had the potential to provide approximate solutions for relatively large problem instances, but its effectiveness was limited to the performance of its local MILP solver. In this paper, we integrate the gossip algorithm with a local solver based on CG, which makes it an effective algorithm for larger problem instances. We also provide numerical experiments and complexity evaluations of the improved gossip algorithm (gossip-CG) with the standard gossip (gossip-MILP) and CG, and show that gossip-CG outperforms the pure CG in case of large problems.

I. INTRODUCTION

The demand for home health care solutions is growing, partly because of the increasing portion of the elderly people in the population, and as a result of economic factors and patient preferences. Manual work plan generation for caretakers is a difficult and time consuming task that should take into account many different constraints such as qualification level of caretakers and their preferred working hours, the required skill level, patients' time windows, and patient's locations. The Home Healthcare Scheduling and Routing Problem (HHCSRP) concerns the optimization of each caretaker's route in a way that an objective specified by the service provider, like travel time and customer satisfaction is minimized. Descriptions of HHCSRP closely resembles the classical vehicle routing problem with time windows (VRPTW), hence VRPTW along with an array of field-specific constraints can be used to model the problem.

A classical vehicle routing problem (VRP) can be stated as the problem of determining an optimal set of routes for a set of vehicles such that given demands at the customers are satisfied. Each route starts and ends in the same depot. Each customer should be visited by just one vehicle and each vehicle has a capacity that can not be exceeded. The routes

are to be selected such that the total cost is minimized [1]. In the time-windowed version of VRP (VRPTW), additional time constraints are present that force the vehicles to visit the customers within a specified time frame.

Due to the importance of the VRP and VRPTW in theory and practice, a plethora of literature are dedicated to the topic. A detailed survey of formulations and exact algorithms for VRPTW is provided in [2]. In another recent survey [3], it is reported that currently the best exact algorithms for VRP and VRPTW are the ones based on column generation (CG) algorithms that solve set-partitioning (SP) models of VRPTW (see ex. [4]). Such algorithms stem from the work of Desrochers et al. [5], where VRPTW was decomposed into a master problem in form of a set-covering (SC) problem, and a subproblem in form of a shortest path problem with time window and resource constraints (SPPTWRC). They combined CG with branching to achieve integral solutions, which resulted in a framework known as branch and price.

In the context of CG for VRP or VRPTW, if the vehicles are assumed to be identical, the problem will be decomposed into one master problem and a set of identical subproblems, requiring only the solution of one subproblem, as in [5]. But in the case of heterogeneous vehicles (HVRP), one would need to solve all of the resulting subproblems. Choi and Tcha [6] employed CG to solve a set-covering model of HVRP with promising results. In this paper, we are interested in an SC model of VRPTW with heterogeneous vehicles (or caretakers) to represent the HHCRSP.

Set-partitioning/set-covering models are popular in the area of home healthcare scheduling. Eveborn et al. [7] utilized an SP model and solved the master problem using a repeated matching algorithm. This is a heuristic approach which has been successfully applied to real-world home care problems. Inspired by the success of the branch and price on SP variants of VRPTW, another study applied the same method to real-life problem instances [8], where temporal dependencies as side constraints during the branching process were enforced.

In another attempt, Riazi et al. [9] heuristically solved HHCRSP using a distributed gossip algorithm. According to the gossip rule, after an initial task assignment, two caretakers and their tasks are randomly picked to form a local optimization problem. The solution to this problem yields either an equal or a better objective function value, through

retaining or exchanging tasks among the caretakers. Then, the same process is repeated for another randomly chosen couple of caretakers, until no further improvement is achieved. Obviously, the performance of this algorithm directly depends on the efficiency of its local solver. The authors used a MILP solver to manage the local problems and were not able to handle very large problems. In this paper, the aim is to turn the gossip algorithm into an effective technique for large HHCRSPs by employing a CG-based local solver.

First, we make use of a standard CG algorithm [5] for solving an SC model of HHCRSP. It was shown in [6] that for an SC model of VRPTW, the optimal value of its linear relaxation is very close to the optimal integer value of the problem, indicating the efficiency of the branch and price as an exact method. To obtain a faster heuristic, we take advantage of this fact, and generate columns only in the root node of the branch and price tree. Since this method does not guarantee integer solutions, we enforce integrality by re-solving the modified master problem as a MILP problem. This approach proves to generate high quality solutions in a reasonable amount of time. The resulting CG algorithm will then serve as the local solver of the gossip algorithm, which will be referred to as gossip-CG.

To summarize the contributions of this paper: (1) we combine gossip algorithm and CG to create gossip-CG algorithm as an alternative to the gossip-MILP for HHCRSP; (2) We present complexity evaluations of the CG and gossip-CG algorithms, as well as the gossip-MILP; (3) We show that the proposed gossip-CG method outperforms the original CG solver, for large problems.

Section II discusses our problem formulation of HHCRSP. Section III provides a background on CG. It treats Dantzig-Wolfe decomposition as a tool to reformulation and preparation of HHCRSP before applying CG. In Section IV gossip algorithm is discussed. Finally, in Section V description of test problems and results of numerical experiments for CG, gossip-MILP and gossip-CG are given.

II. PROBLEM DEFINITION

To describe the HHCRSP, we use a general MILP formulation of VRPTW [10], and add a few side constraints to make it more fitting to healthcare context. It is noteworthy that our model is a simplified and theoretical formulation that does not capture all of the real-world constraints. Yet, it allows us to focus on the core requirements of an actual HHCRSP and to test our algorithms for them. Later in Section III, this model will be converted to a set-covering type, suitable for CG algorithm.

The problem is modeled using a graph. There is a set of n nurses (or caretakers) $\mathcal{N}$, and a set of p patients $\mathcal{P}$ represented by nodes $1, 2, \dots, p$. All caretakers start from the same depot, and return to it after completing their jobs. For the sake of convenience, the depot is modeled by two identical nodes in the graph, one for starting from, and the other for ending to. Patients are however scattered in the region and a node is assigned to each of them. The problem is then defined

by the sets $\mathcal{N}$, $\mathcal{P}$, and a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, where $\mathcal{V}$ is the set of all nodes $0, 1, \dots, p+1$, and $\mathcal{A}$ is the set of arcs. Each arc $(i, j) \in \mathcal{A}$ represents a direct connection between nodes $i, j \in \mathcal{V}$. There are no arcs originating from node $p+1$ or going to node 0, and there are no self arcs, i.e. $(i, i) \notin \mathcal{A}$. The following sets of variables are also defined:

- $\mathcal{X}$: The set of binary sequencing variables x_{ijk} , where $(i, j) \in \mathcal{A}$ and $k \in \mathcal{N}$. If $x_{ijk} = 1$, the caretaker k visits patient j directly after patient i
- $\mathcal{T}$: The set of continuous timing variables t_{ik} defined for each node $i \in \mathcal{V}$ and each caretaker $k \in \mathcal{N}$. If patient i is to be visited by caretaker k , the value of t_{ik} determines when this should happen, otherwise its value has no meaning.

The other constants and costs are as follow:

- Caretaker's time window $[\bar{a}_k, \bar{b}_k]$: $\bar{a}_k$ and $\bar{b}_k$ are the starting and finishing time for the work-shift of caretaker $k \in \mathcal{N}$.
- Patient's time window $[\bar{a}_i, \bar{b}_i]$: $\bar{a}_i$ and $\bar{b}_i$ are the earliest time and the deadline for serving patient $i \in \mathcal{P}$.
- Caretaker's qualification level q_k for caretaker $k \in \mathcal{N}$.
- Patient's qualification requirement r_i for patients $i \in \mathcal{P}$.
- Duration of the patient's demanded service $\bar{t}_i$: The amount of time a caretaker needs to spend serving patient $i \in \mathcal{P}$.
- The symmetric cost matrix C : It contains the Euclidean distance c_{ij} defined for every arc in $\mathcal{A}$.
- Travel time $\bar{t}_{ij}$: This value is calculated by dividing the euclidean distance between nodes i and j by caretaker's travel speed.

Therefore, the MILP formulation could be given as:

$$\min z = \sum_{k \in \mathcal{N}} \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{V}} c_{ij} x_{ijk} \quad (1)$$

$$\sum_{k \in \mathcal{N}} \sum_{j \in \mathcal{V}} x_{ijk} = 1 \quad \forall i \in \mathcal{P} \quad (2)$$

$$r_i x_{ijk} \leq q_k \quad \forall i \in \mathcal{P}, \forall j \in \mathcal{V}, \forall k \in \mathcal{N} \quad (3)$$

$$\sum_{j \in \mathcal{P}} x_{0jk} = 1 \quad \forall k \in \mathcal{N} \quad (4)$$

$$\sum_{i \in \mathcal{P}} x_{ijk} - \sum_{i \in \mathcal{P}} x_{jik} = 0 \quad \forall j \in \mathcal{P}, \forall k \in \mathcal{N} \quad (5)$$

$$\sum_{i \in \mathcal{V}} x_{i,p+1,k} = 1 \quad \forall k \in \mathcal{N} \quad (6)$$

$$t_{ik} + \bar{t}_{ij} + \bar{t}_i - M_{ij}(1 - x_{ijk}) \leq t_{jk} \quad \forall i, j \in \mathcal{V}, \forall k \in \mathcal{N} \quad (7)$$

$$\bar{a}_i \leq t_{ik} + \bar{t}_i \leq \bar{b}_i \quad \forall i \in \mathcal{P}, \forall k \in \mathcal{N} \quad (8)$$

$$\bar{a}_k \leq t_{ik} \leq \bar{b}_k \quad \forall i \in \mathcal{V}, \forall k \in \mathcal{N} \quad (9)$$

$$x_{ijk} \in \{0, 1\} \quad \forall i, j \in \mathcal{V}, \forall k \in \mathcal{N} \quad (10)$$

$$t_{ik} \geq 0, t_{ik} \in \mathbb{R} \quad \forall i \in \mathcal{V}, \forall k \in \mathcal{N} \quad (11)$$

The objective function given by (1) minimizes the total distance travelled by the caretakers. Constraints in (2) assure

that any patient is visited once, and by only one caretaker. Constraints in (3) assures that only a qualified enough caretaker is assigned to any job. Constraints (4), (5), and (6) assert that each caretaker leaves the depot; after a caretaker visits a patient he/she must leave for another location; all caretakers must comeback to depot (node $p+1$). Constraints in (7) states the relationship between the total spent time for a patient before leaving them, and the next patient's service start time. Note that the large constant M_{ij} can be reduced to $\max\{\bar{b}_i + \bar{t}_{ij} - \bar{a}_j\}$, $(i, j) \in A$. Furthermore, (8) affirms that patients are visited within their time windows, while (9) guarantees that any caretaker works within his/her specified work hours. Finally, (10) and (11) define the domains of the variables.

Notice that in the above formulation, there is no need to include sub-tour elimination constraints, in contrast to VRP. This is due to the presence of the constraints containing service time variables t_{ik} that enforce a unique route direction. Moreover, for the sake of simplicity, it is assumed that no patient can be visited by more than one caretaker, as opposed to more realistic models as in [7].

III. COLUMN GENERATION

In the heart of many CG algorithms there is a decomposition mechanism such as Dantzig-Wolfe decomposition (DWD) [11]. The idea behind any decomposition approach is to take advantage of the problem structure to facilitate the solution procedure. This is done by partitioning the problem into smaller and easier-to-handle segments that are solved separately, and by letting those segments communicate with each other. DWD is one such method that was first developed to operate on linear programming (LP) problems with block-angular constraint matrix. It is possible to apply DWD to structured MILP problems as well (see [12]), and this will happen in Section III-B to the HHCRSP.

A. Dantzig-Wolfe decomposition

Consider an LP problem with a combination of easy constraints and complicating (coupling) ones, i.e. those involving variables coupled together with non-zero coefficients. First, the coupling (complicating) constraints should be identified. Then, the problem is *reformulated*, such that a feasible solution is now represented by a convex combination of the extreme points of the polyhedron defined by the easy constraints (non-complicating ones). The reformulated problem is now called the *master problem* (MP), and it involves fewer constraints, but it has considerably more variables that makes it difficult to construct, and difficult to solve by simplex algorithm. Since in every iteration of simplex algorithm only a few of the variables are present in the basis, we choose to construct and start the master problem based on only a meaningful subset of the extreme points (variables). This is known as the *restricted master problem* (RMP). To improve the objective function value of RMP, a new variable should enter the basis. The new variable is identified through solving one (or more) auxiliary problem(s) known as the *subproblem(s)*. In other words, any

variable in the RMP corresponds to a solution to one of the subproblems.

Subproblem is defined as the problem of finding a variable that improves the objective function of the RMP the most, such that the easy constraints are satisfied. Hence, the objective function value of the subproblem is the *reduced cost* of a variable entering the basis. The communication of the RMP with the subproblem is done through manipulating the objective function of the subproblem by using solution values of the dual variables corresponding to RMP. If the reduced cost is negative (in case of a minimization problem), a new variable is added to RMP (as a new column) with its corresponding coefficients. This process continues until no new variable can be identified. Therefore, we can see that instead of solving the original problem, we solve a restricted master problem, and one (or more) subproblem(s), while these smaller problems exchange information. For a more detailed treatment of the topic, see [12].

As mentioned earlier, DWD can also be applied to structured MILPs. In this case, the CG scheme will be slightly different in the sense that instead of solving a MILP master problem, its LP relaxation is solved. This implies that the optimal solution should be checked for integrality. If the solution is integral, it is then the optimal integer solution of the problem. Otherwise, a branch and bound procedure should be employed such that in every node of the resulting tree CG algorithm is executed to generate more columns. The combination of branch and bound and CG is known as *branch and price*. For more information on different branching strategies for VRPTW, the reader is referred to [10] and [13]. A branching scheme for a model with variable number of columns cannot be implemented using commercial solvers such as CPLEX, but there are other platforms like ABACUS [14] to facilitate the branching. Instead of the branch and price method, one could opt for a heuristic version of CG, where after solving the root node (of a branch and price tree) to optimality, the master problem including its new variables is solved once again as a MILP to obtain integer solution. We choose to employ such a heuristic CG in this paper.

B. Master problem: route assignment

We now give more detail on the construction of MP in the context of the CG. Consider the constraints (2)-(9) in our HHCRSP formulation. It is clear that only the constraints in (3) are coupling the caretakers together, that is, for every $i \in \mathcal{P}$ variables belonging to different caretakers (x_{ijk} , $k \in \mathcal{N}$) are present in a row. We refer to (3) as the difficult constraints, and call (4) through (9) the easy ones. If we set the difficult constraints aside, the easy ones are simply separated as a set of individual subproblems in form of elementary shortest path problems with time window and qualification constraints (ESPTWQC). Therefore, for every caretaker, there will be an ESPTWQC. A polytope defined by all the easy constraints related to a caretaker has a set of extreme points, and each extreme point attributes to a valid path for caretaker $k \in \mathcal{N}$. Each set of extreme points is denoted as Ω_k , $k \in \mathcal{N}$,

and each extreme point (valid path or route) is denoted as λ_k^r , $r \in \Omega_k$, $\forall k \in \mathcal{N}$

Now, as mentioned in Section III-A, after identifying the easy and complicating constraints, the problem should be reformulated. According to definition of convex combination (see [15]), any arc x_{ijk} , $k \in \mathcal{N}$ can be re-written as a convex combination of the extreme points of Ω_k , $k \in \mathcal{N}$, as follows:

$$x_{ijk} = \sum_{r \in \Omega_k} x_{ijk}^r \lambda_k^r \quad \forall k \in \mathcal{N} \quad \forall (i, j) \in \mathcal{A} \quad (12)$$

$$\sum_{r \in \Omega_k} \lambda_k^r = 1 \quad \forall k \in \mathcal{N} \quad (13)$$

$$\lambda_k^r \geq 0 \quad k \in \mathcal{N} \quad \forall r \in \Omega_k \quad (14)$$

Note that x_{ijk}^r is nothing but a coefficient whose values is 1 if arc (i, j) belongs to the path related to λ_k^r , and 0 otherwise. We also define c_k^r as the cost of route r for caretaker k , and a_{ik}^r as the number of times caretaker k visits a patient i , as follows:

$$c_k^r = \sum_{(i, j) \in \mathcal{A}} c_{ij} x_{ijk}^r \quad k \in \mathcal{N} \quad \forall r \in \Omega_k \quad (15)$$

$$a_{ik}^r = \sum_{j \in \mathcal{P}} x_{ijk}^r \quad \forall i \in \mathcal{V} \quad k \in \mathcal{N} \quad \forall r \in \Omega_k \quad (16)$$

Reformulating the HHCRSP using the path variables results in the following master problem:

$$\min \sum_{k \in \mathcal{N}} \sum_{r \in \Omega_k} c_k^r \lambda_k^r \quad (17)$$

$$\sum_{k \in \mathcal{N}} \sum_{r \in \Omega_k} a_{ik}^r \lambda_k^r = 1 \quad \forall i \in \mathcal{V} \quad (18)$$

$$\sum_{r \in \Omega_k} \lambda_k^r = 1 \quad \forall k \in \mathcal{N} \quad (19)$$

$$\lambda_k^r \geq 0 \quad k \in \mathcal{N} \quad \forall r \in \Omega_k \quad (20)$$

So, the MP is expressed in path formulation, as opposed to the original problem stated in arc formulation. It also resembles a set-partitioning problem. If the equality in (18) is replaced by inequality, MP becomes a set-covering type of problem. For practical reasons such as numerical stability, a set-covering type is superior to a set-partitioning one. Note that, if the cost matrix satisfies the triangular inequality, over-covering never occurs, i.e. a patient is never visited more than once.

As stated in Section III-A, the MP includes many variables and it is practically difficult to solve. The CG approach attacks on a restricted MP with only a few variables, and adds the required variables iteratively. We warm-start the RMP with a feasible path for each caretaker, and the feasible paths are obtained heuristically from the original problem formulation.

C. Subproblems: constrained shortest path problems

Each subproblem is form of an elementary shortest path problems with time window and qualification constraints (ESPTWQC). We choose to separate the qualification constraints (3) from the rest and check them before solving a

subproblem. That is, any patient that cannot be served by a caretaker is not included in the subproblem particular to that caretaker, hence the subproblem becomes a shortest path problem with time window (ESPTW).

As previously mentioned in Section III-A, the role of the subproblem is to determine which new variable could improve the objective function value of the RMP the most. In our case, the task is identifying a new route pertaining to a caretaker that helps reducing the total travel cost. Hence, a negative reduced cost for that variables is desirable. Given that, the objective function (reduced cost) for any subproblem $k \in \mathcal{N}$ is given by

$$\min \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{V}} (c_{ij} - \pi_i) x_{ijk} - \pi_k$$

where π_i and π_k are the values of the dual variables corresponding to the i^{th} and k^{th} constraint in (18) and (19) respectively.

The subproblems are NP-hard and there is no polynomial algorithm to solve them. But we use the method suggested by [5] that allows us to solve a relaxation to ESPTW in pseudo-polynomial time using a label-setting algorithm.

IV. GOSSIP ALGORITHM

The gossip algorithm, which is an example of distributed algorithms, is popular in telecommunication and sensor networks for exchanging information and for computing in an arbitrarily connected network of nodes (see for example [16]). In a gossip algorithm, every node communicates with just one other neighboring node. Therefore, the computations are not done in a centralized fashion.

Gossip approach for heterogeneous multi-vehicle routing problem (HMVRP) was first implemented in [17], where the vehicles mimic the sensors in a sensor network and communicate with each other in a pairwise fashion, while exchanging information about the optimal way of doing their sets of tasks. According to the gossip rule, after an initial task assignment, two vehicles and their tasks are randomly picked to form a local optimization problem. The solution to this problem yields either an equal or a better objective function value by retaining or changing the task assignments among the vehicles. Then, the same process is repeated for another randomly chosen couple of vehicles, until no further improvement is achieved.

For example, consider a small HMVRP instance with 3 vehicles and 9 tasks, as shown in Figure 1, where each cluster of vehicle-tasks is in the same color. In the first iteration (upper-left corner), an arbitrary feasible task assignment is as follows: (tasks 1, 2, 3 $\rightarrow$ vehicle A), (tasks 4, 5, 8 $\rightarrow$ vehicle B), (tasks 6, 7, 9 $\rightarrow$ vehicle C). Then two vehicles and their corresponding tasks are chosen randomly, say vehicle A and B, and they form a local optimization problem. Optimizing this local problem results in task 4 to be passed to vehicle A, as this decreases the objective function of the local problem. Next, vehicles B and C are chosen, and task 9 is transferred from C to B. Then, Vehicles A and C are picked, which yields no new results, as they are already optimal with regard to

each other. This process is shown in Figure 1. Notice that the method has no specific stopping condition. The important rationale behind the algorithm is that it allows to solve local problems that are easier to solve, compared to a monolithic (centralized) approach.

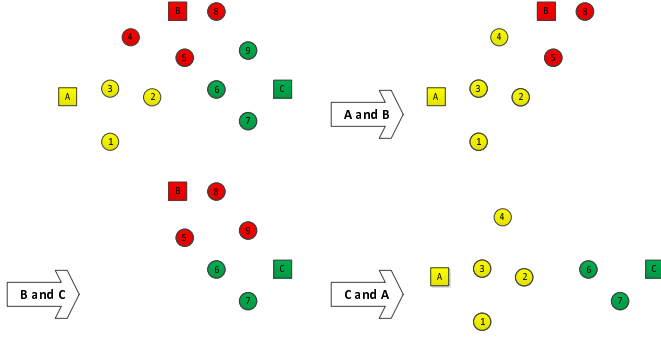


Fig. 1. Some iterations of gossip algorithm for a small instance of HMVRP.

Gossip and its extended version were used for solving the HHCRSP in [9], with a standard MILP solver for the local problems. Since the performance of a gossip algorithm is depending on that of its local solver, gossip-MILP becomes ineffective on large problems. For example, in healthcare context, when the service time for patients is relatively short, the corresponding caretaker can attend to more patients, and therefore he/she will have a longer list of tasks to be scheduled. In this case, optimizing the tasks assignment and scheduling the tasks for a pair of such caretakers in a local problem of gossip algorithm may be troublesome. Next, we will improve the gossip by using a CG-based solver for the local problems and show that the resulting algorithm (gossip-CG) outperforms the standard CG.

V. NUMERICAL EXPERIMENTS

The MILP solver used in gossip-MILP was IBM ILOG CPLEX 12.5 (64-bit). The algorithms were implemented in C++ using Microsoft Visual Studio in Microsoft Windows 7 Enterprise environment. The experiments were conducted on a PC with an Intel Core2 Quad CPU (2.66 GHz), and 8 GBs of RAM.

A. Implementation details

Both the gossip-MILP and the gossip-CG algorithms should be started with an initial (preferably feasible) solution. We adopted an insertion heuristic as in [18], which was able to find a feasible solution for a problem of 150 nodes in less than a second. The CG algorithm is also warm-started using the same heuristic. CG subproblems were solved using a label-setting algorithm with 2-cycle elimination [5]. We tuned the gossip-MILP algorithm such that the underlying MILP solver was not allowed to spend more than a specified amount of time (120 seconds) on any local problem. This prevented the algorithm from getting stuck in a difficult local problem.

B. Problem instances

We have evaluated the CG and gossip methods for different configurations of caretakers and patients. Each configuration is indicated by a tuple (n, p, i) to indicate the number of caretakers, patients, and the problem instance number. Our small problems comprise of 3 caretakers and 25 patients. The medium problems include 6 to 8 caretakers, and 40 to 60 patients. The large problems have 8 to 10 caretakers and 90 to 100 patients. For each of these combinations, the instances were generated randomly within specific bounds simulating the real-life conditions. Caretakers may leave the depot at 8:00 and should be back by 16:00. They spend between 10 to 20 minutes on each patient (service time), and all travel at the same speed, 17 km/h. The patients are randomly located in an area of $5 \times 5 \text{ km}^2$, and admit the caretakers in caretakers' working hours. They have a time window between 30 to 120 minutes for their required service. The patients have demands for qualification level of the caretakers, ranging from 1 (basic skills required) to 4 (specialists needed). The caretakers, on the other hand, offer qualification level from 2 (average) to 5 (professional).

Since local problems in gossip methods are chosen randomly, different runs demonstrate somewhat different behaviors of the gossip. Each problem instance was solved 10 times for the gossip methods to obtain a better picture of their average performance in terms of solution time and best integer solution obtained. For the gossip algorithms, the solution time is reported as the latest time at which an improvement in the solution is seen, and the problem instances were solved under different time limitations. However, for the CG algorithm, a fixed time limit of 2 hours was used.

C. Results and discussion

The results of the experiments are given in Table I. It is seen that for the small problems, i.e. (3, 25, 1-3) gossip-MILP gives better integer solutions compared to both CG and gossip-CG. This is not surprising, since the local solver of the gossip-MILP yields exact solutions, compared to that of gossip-CG, which uses a heuristic version of CG. The average solution times for both gossip-MILP and gossip-CG are well below the defined time-limit, which is an indication of possible convergence of the algorithms to the reported bounds. Also, the small standard deviations of the bounds hint at possible convergence.

For medium-sized problems (6, 40, 1-3), (7, 50, 1-3) and (8, 60, 1-3), it is observed that CG gives similar bounds to gossip-MILP and gossip-CG, under the time limit imposed to the gossip algorithms, i.e. under 300, 600 and 900 (s). Both gossip algorithms demonstrate similar behaviors to each other.

For large problems, CG fails to terminate before the time limit of 7200 (s). After 2 hours it finishes with LP bounds of 73.2 and 75.5 for (8, 90, 1) and (10, 100, 1) respectively. Gossip-MILP, however, continues its operation, thanks to the mentioned modification in its local solver (Section V-A). The gossip-CG, on the other hand, yields slightly better integer bounds. Although its local solver (CG) is not exact and

TABLE I

COMPARISON OF CG, GOSSIP-MILP AND GOSSIP-CG: n , p AND i ARE THE NUMBER OF CARETAKERS, NUMBER OF PATIENTS, AND THE PROBLEM INSTANCE NUMBER. EMPTY CELLS INDICATE THAT SOLUTION WAS NOT AVAILABLE. *TiLim* AND *SolTi* DENOTE THE IMPOSED TIME LIMIT AND SOLUTION TIME IN SECONDS. *Sol* DENOTES THE BEST FOUND INTEGER SOLUTION. *SolTiAve* IS THE AVERAGE SOLUTION TIME AT WHICH THE BEST INTEGER WAS FOUND, OVER 10 RUNS, FOR THE SAME INSTANCE. *TiStd*, *SolAve* AND *SolStd* DENOTE THE STANDARD DEVIATION OF THE SOLUTION TIME, THE AVERAGE SOLUTION, AND THE STANDARD DEVIATION OF THE BEST INTEGER SOLUTION, OVER 10 RUNS FOR THE SAME INSTANCE

size			CG			gossip-MILP					gossip-CG				
n	p	i	TiLim	SolTi	Sol	TiLim	SolTiAve	TiStd	SolAve	SolStd	TiLim	SolTiAve	TiStd	SolAve	SolStd
3	25	1	7200	64.6	33.7	300	113.1	78.6	30.2	0.6	300	23.6	13.6	34.5	0.2
		2	7200	36.1	50.5	300	59.9	41.8	43.6	0.0	300	12.1	7.5	53.8	0.0
		3	7200	56.6	47.3	300	19.5	11.9	39.8	0.0	300	10.1	7.08	43.1	3.7
6	40	1	7200	159.0	51.9	300	261.4	5.1	47.3	4.3	300	105.8	23.8	52.8	3.1
		2	7200	170.4	46.9	300	218.1	62.3	44.9	3.4	300	204.7	56.9	45.9	3.1
		3	7200	83.1	59.3	300	248.2	27.7	62.7	3.7	300	180.5	70.2	63.2	3.3
7	50	1	7200	476.8	60.4	600	455.4	111.2	58.8	5.5	600	307.3	106.9	56.7	2.2
		2	7200	222.3	59.1	600	472.1	126.1	55.6	1.5	600	374.5	137.1	60.6	4.5
		3	7200	221.4	53.8	600	446.6	154.6	54.4	1.1	600	401.8	94.3	55.5	3.1
8	60	1	7200	957.3	62.5	900	806.0	79.3	58.0	1.7	900	741.9	145.6	65.3	2.8
		2	7200	715.5	65.3	900	740.5	147.1	66.3	1.3	900	807.1	42.7	68.6	1.1
		3	7200	891.3	75.2	900	726.9	159.1	87.5	10.1	900	691.9	151.1	77.5	3.1
8	90	1	7200			1500	1340.1	114.9	139.4	7.0	1500	1387.5	109.8	134.7	11.1
10	100	1	7200			1500	1329.4	80.8	126.9	13.8	1500	1296.8	216.4	116.5	4.8

yields weaker bounds compared to the MILP solver in gossip-CG, it solves local problems much faster. Hence, gossip-CG goes through more iterations than gossip-MILP does, and that gives it a better chance of improving the bounds by pairing more caretakers within the time limit, compared to the slower gossip-MILP.

VI. CONCLUSIONS

We have presented the gossip-CG algorithm for HHCRSP, and have compared its performance with gossip-MILP and pure CG. The results of the numerical experiments indicate that gossip-CG outperforms pure CG for large problems. The subproblems in CG were tackled using a standard method, without paying much heed to optimizing its performance. Nevertheless, the resulting gossip-CG algorithm showed similar performance to our well-tuned gossip-MILP. On large problems, gossip-CG provided better bounds compared to gossip-MILP, and it is likely that it can outperform gossip-MILP on medium-sized problems as well, if a more efficient algorithm is used to handle the CG subproblems.

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